

OEE Process with Sensors

How Overall Equipment Effectiveness is measured using machine sensors / PLC signals — and how that process maps to LineSight for beverage filling lines.

OEE = Availability × Performance × Quality

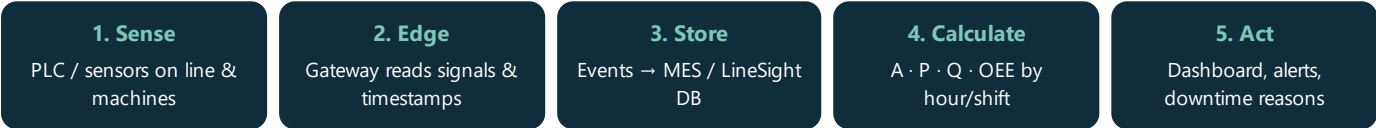
1. Availability
Run Time ÷ Planned Time
Is the line running when it should be?
Losses = stops, breakdowns, waiting, changeover.

2. Performance
(Ideal Cycle × Count) ÷ Run Time
Is the line running at the designed speed? Losses = slow cycles, micro-stops.

3. Quality
Good Count ÷ Total Count
Are bottles/cases within specification?
Losses = rejects, rework, scrap.

End-to-end process (sensor-based OEE)

Data flows left → right every second (or every machine cycle)



What each sensor / signal measures

OEE pillar	Typical sensors / signals	What you get
Availability	Machine Run/Stop bit (PLC) Stack light / status light tap Current (amp) sensor Vibration / proximity for idle vs run	Exact start/stop of downtime Micro-stops under 1 minute Run Time vs Planned Time
Performance	Photo-eye / bottle counter Cycle complete pulse from PLC Filler / blower speed (BPM/CPH)	Actual cases or bottles per hour Speed loss vs ideal rate Short stops that hurt output
Quality	Reject chute counter Vision inspection (fill/cap/label) Checkweigher / leak detector	Good vs reject count Defect type (optional) Quality % for the hour/shift

Important: Sensors give *when* and *how much*. Operators / supervisors still enter *why* (downtime reason, changeover type). LineSight already stores reasons; sensors automate the times and counts.

From sensors to LineSight screens

Recommended process for a beverage plant (Nakshatra-style filling line).

Step-by-step implementation process

1. **Define Planned Time** — from Work Order Planning (shift start/end, planned operating minutes).
2. **Wire Availability signals** — Run/Stop from main line PLC or current sensor on critical machines (Filler, Capper, Blow Mould).
3. **Wire Performance counters** — case/bottle count at outfeed; set Ideal Cycle Time / Cases per Hour in Production Targets.
4. **Wire Quality counters** — good count + reject count (or total – rejects).
5. **Edge gateway** — poll every 1 second; push events to API / database.
6. **Map to LineSight** — auto-create hourly Production Entries + Downtime start/end; operator confirms reason.
7. **Validate 2 weeks** — compare sensor OEE vs manual Production Entries until they match.
8. **Go live** — dashboards (Plant, Line-wise, Day/Week OEE) use live data; keep manual entry as backup.

How it maps to your current LineSight app

Today (manual)	With sensors (future / hybrid)
• Work Order Planning → date, line, shift, times • Production Entries → hourly planned / actual / good / loss • Downtime Entries → start, end, machine, reason • Changeover Details → type, from/to SKU, minutes • Dashboards calculate $A \times P \times Q$	• Same work orders (planned time stays master) • Counters auto-fill actual / good / reject • Run/Stop auto-fills downtime windows • Operator only picks reason / category • Same OEE formulas — cleaner data

Simple example (1 hour on LINE-01)

Input	Value	Source
Planned hour	60 min	Work order
Downtime (unplanned)	10 min	Run/Stop sensor + reason code
Run Time	50 min	$60 - 10$
Ideal rate	180 cases/hour	SKU / Production Target
Actual cases	135	Counter sensor
Good cases	130	Counter – rejects / vision
Availability	$50 \div 60 = \mathbf{83.3\%}$	Formula
Performance	$(180 \times (50/60)) \text{ vs } 135 \rightarrow \sim \mathbf{90\%}^*$	Formula
Quality	$130 \div 135 = \mathbf{96.3\%}$	Formula
OEE	$0.833 \times 0.90 \times 0.963 \approx \mathbf{72\%}$	$A \times P \times Q$

*Performance uses Ideal Cycle Time × Total Count ÷ Run Time (LineSight uses the same pillar logic as Plant / Line / Day OEE).

Recommended sensor set for a filling line (starter)

- **Must-have:** Line Run/Stop (PLC or current sensor) · Outfeed case/bottle counter · Reject counter
- **Nice-to-have:** Per-machine status (Filler, Capper, Labeller) · Vision for quality · Changeover detection from recipe/SKU change
- **Always keep:** Operator downtime reason entry in LineSight (sensors cannot know “why” alone)

Practical tip: Start with *one line* and *Availability + count* only. Add quality sensors after counts match manual Production Entries for 2 weeks.

